Slide 11: the listener

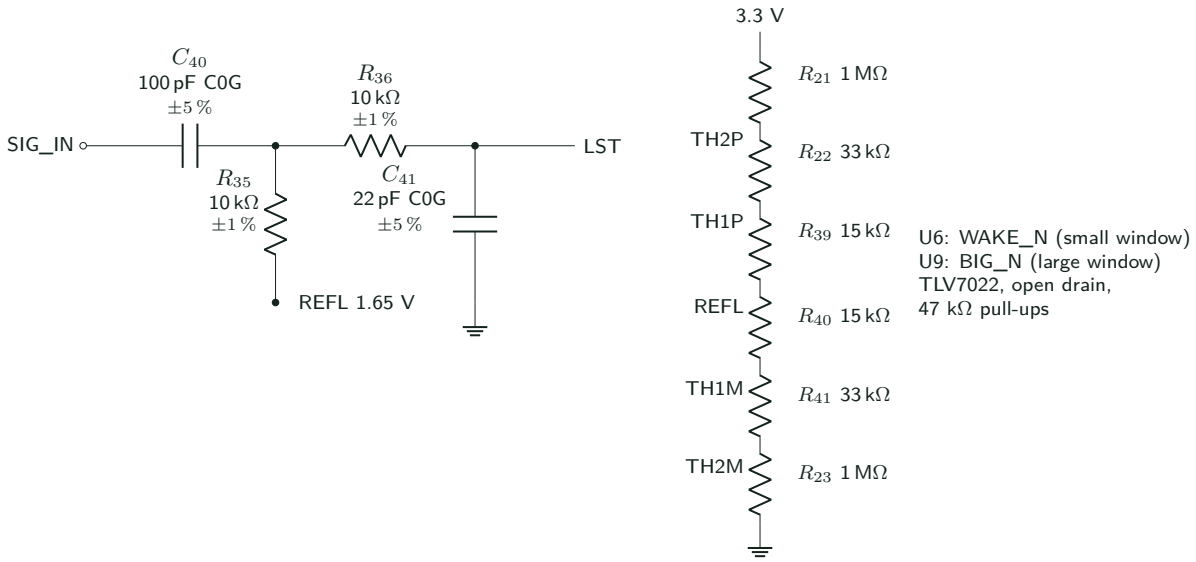
WAKE_N and BIG_N watch the signal without gating it

ADS7057 test board v3 — study notes

1 The idea

The listener is a small second circuit that reads SIG_IN, the same signal the chain sees. It never switches the capture chain: the chain stays on and the FPGA has the waveform and the listener decisions side by side. Two open-drain lines go to the PMOD: WAKE_N (medium burst, PMOD pin 10) and BIG_N (large burst, pin 7). Both are active low.

2 The circuit



3 The filter

C40/R35 is a high-pass with its corner at $1/(2\pi \cdot 10\,\mathrm{k\Omega} \cdot 100\,\mathrm{pF}) = 159\,\mathrm{kHz}$. R36/C41 is a low-pass at $1/(2\pi \cdot 10\,\mathrm{k\Omega} \cdot 22\,\mathrm{pF}) = 723\,\mathrm{kHz}$. R35 returns to REFL, so the output rests at 1.65 V. The second stage loads the first, so the exact gain from the two node equations (not the product of two corners) is

f	10 kHz	100 kHz	250 kHz	1 MHz
$\lvert H\rvert$	0.063	0.489	0.681	0.529
dB	−24.1	−6.2	−3.3	−5.5

ngspice on the as-built netlist gives −3.34 dB at 250 kHz and −24.06 dB at 10 kHz, the same numbers. The netlist simulation found that the first version of this filter loaded its first stage too much. R36 and C41 were changed to 10 kΩ and 22 pF to fix it.

4 The thresholds

One chain of resistors from 3.3 V to ground makes all four thresholds: R21 1 M, R22 33 k, R39 15 k, **REFL**, R40 15 k, R41 33 k, R23 1 M. Total 2.096 MΩ, so

$$I = \frac{3.3}{2.096\,\mathrm{M\Omega}} = 1.574\,\mu\mathrm{A}, \qquad V_{\mathrm{REFL}} = 3.3 \cdot \frac{1.048}{2.096} = 1.65\,\mathrm{V}.$$

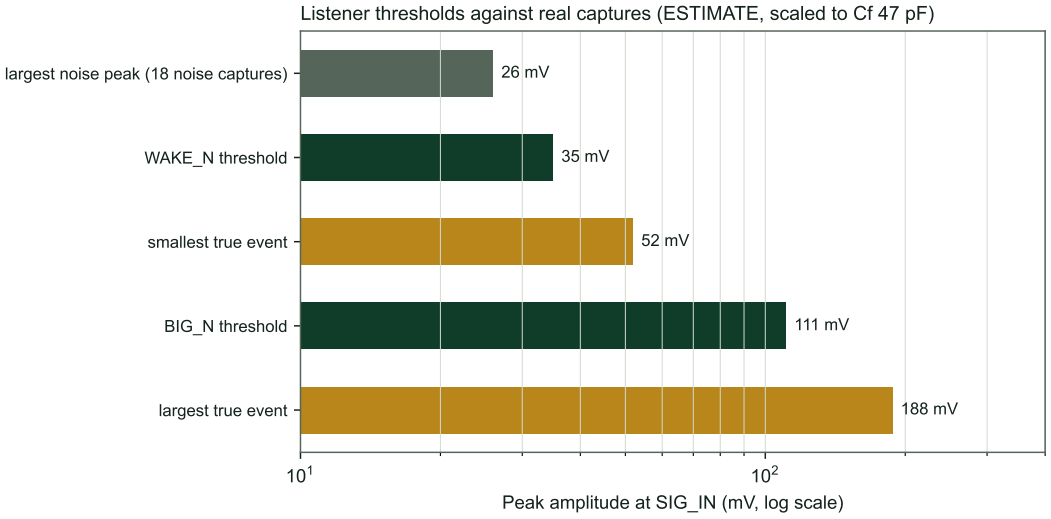
$$V_{\mathrm{TH1}} - V_{\mathrm{REFL}} = I \cdot 15\,\mathrm{k\Omega} = 23.6\,\mathrm{mV}, \qquad V_{\mathrm{TH2}} - V_{\mathrm{REFL}} = I \cdot 48\,\mathrm{k\Omega} = 75.6\,\mathrm{mV}.$$

Each comparator pair (U6 for WAKE_N, U9 for BIG_N) compares LST with a node above and a node below REFL, so either polarity trips it. Divide by the filter gain 0.681 to refer the thresholds to SIG_IN:

$$\text{WAKE_N: } \frac{23.6}{0.681} = 35\,\mathrm{mV}, \qquad \text{BIG_N: } \frac{75.6}{0.681} = 111\,\mathrm{mV}.$$

The 47 kΩ pull-ups (R24, R42) make the lines open drain: the TLV7022 can only pull low. Response time about 260 ns, input offset up to ±8 mV and hysteresis about 7 mV, so each threshold is good to about ±12 mV at SIG_IN.

5 Against real data



On 25 recorded captures (scaled to $C_f = 47$ pF, an estimate) the noise after the 100 kHz high-pass is 4.0 mV rms (worst 9.2 mV). The largest noise peak of the 18 noise captures is 26 mV, under the 35 mV WAKE_N threshold, a margin of only 1.3×. The 7 real events peak at 52 to 188 mV, all above it.

> The margin on noise is thin, and the thresholds move with the head. With the OPA314 head ($C_f = 220$ pF) volts per pC fall by 4.7 and the ladder must change. Bench test: 250 kHz ten-cycle burst; WAKE_N must fall near 35 mV and BIG_N near 111 mV at SIG_IN.